

Nico Seamans

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EDUCATION

B.Sc. in Mechanical Engineering (25-26 Dean's GPA List), Minor in Mathematics

University of New Brunswick

Dec 2027 graduation

Fredericton, New Brunswick

WORK EXPERIENCE

Aerospace Manufacturing Engineering Intern

Pratt & Whitney

Jan 2026 – Aug 2026

Enfield, Nova Scotia

- Designed and built an airfoil lab fixture for CMM inspection to validate machining tolerance in accordance with GD&T specifications, reducing setup-induced measurement error and improving inspection speed and accuracy by ~10%
- Designed 3D-printed polishing templates in CATIA V5 to support Design for Manufacturing (DFM), accelerating manual polishing on IBF fan blades by ~25% (trailing edge) and 14% (leading edge)
- Developed manufacturing processes and documentation, including process flows, machine utilization, and basic NC programming, for the Military Collaborative Combat Aircraft (CCA) Project; led meetings and presented updates to engineers and senior managers
- Partnered with engineers, fellows, and senior managers to specify and source a 5-axis mill-turn machine for projected production volume increases and new program requirements

Co-Founder & Senior Project Lead

UNB Formula SAE EV

Oct 2025 – Present

Fredericton, New Brunswick

- Co-leading the dynamics team of 30+ students in the Formula SAE EV program, coordinating cross-functional timelines toward the team's first competition
- Leading Manufacturing for the team, directing fixture and jig design and establishing process standards and work instructions for custom part machining to enable high-precision, repeatable manufacturing of critical components
- Built early quarter- and half-car models in MATLAB/Simscape to guide suspension geometry and validate handling targets
- Co-designed the rear hub and tripod housing for manufacturability, controlling critical tolerances through GD&T and validating safety factors against design requirements with ANSYS FEA, reducing production time and cost

Mechanical Engineering Student

J.D. Irving Limited

May 2025 – Dec 2025

Chipman, New Brunswick

- Led design and installation of a remote-controlled gantry crane spanning seven log beds, applying root-cause analysis to cut operator exposure and reduce maintenance downtime
- Designed a two-stage winch recovery system to extract trapped logs from woodchipper operations, implementing a corrective action that removed the need for manual entry
- Managed design and installation of a retrofit fall-protection system, including a crossbeam tie-off structure, improving safety compliance for millwright operations

TECHNICAL SKILLS

CAD & Engineering: CATIA V5, Siemens NX, SolidWorks, Fusion 360, Technical Drawings, FEA (ANSYS)

Manufacturing: DFM, GD&T, Fixture Design, FAI, Lean Six Sigma White Belt, Manufacturing Processes, SAP, Enovia

Analysis & Programming: MATLAB, Simulink, Simscape, Excel, Python

EXTRACURRICULARS

- Team Canada Powerlifter: 2025 World Bench Press Bronze Medallist (Drammen, Norway); 2024 Best U24 Bench Presser in Canada; 2025 New Brunswick Powerlifting Best U24 Powerlifter
- NBPL Chair & Board Member
- 2024 Atlantic Engineering Competition: Junior Design (presented design to judges)

PROFESSIONAL PROJECT

Airfoil Lab Fixture

- **Problem:** Blades needed to be securely and consistently fastened to enable fast, repeatable CMM measurements of the airfoil
- **Solution:** Designed a magnetic bridge fixture with 7 points of contact, secured by a thumb screw to maintain consistent pressure and location across measurements, per GD&T inspection requirements

ACADEMIC PROJECT

Dog Ball Launcher

- **Problem:** Create an indoor tennis ball launcher for a hallway
- **Solution:** Designed a dual-flywheel launching mechanism in CAD, developing the system from concept through detailed part drawings and component selection

PASSION PROJECT

Indoor Golf Simulator

- **Problem:** Create an indoor retractable golf simulator on a limited budget
- **Solution:** Designed and built a motorized retractable impact screen, including the mechanical drive system and remote-control mechanism
- **Drive Train:** The drive system is designed around a reversible 12 V worm-gear motor, chosen for its high torque in a small package. The motor shaft is 8 mm, matching the 15-tooth sprocket, which drives a 32-tooth sprocket through #25 roller chain. That gives about a 2.1:1 additional reduction, trading speed for torque so the screen lifts smoothly. A 12 V, 5 A supply powers the motor, and a wireless relay module with a 50 m range switches its direction, so the screen raises and lowers from a remote with no wiring to the user.